QPCODE **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) THEORY EXAMINATION MARCH 2026
FOURTH SEMESTER
DSE–MG4DSEPHY204–MATERIAL CHARACTERIZATION TECHNIQUES
(2024 ADMISSION ONWARDS)

Duration: 1.5 HOURS **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) andAppreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1.	Discuss Bragg's law with a neat diagram.	U / CO1
2.	What is Laue Method in X-ray spectroscopy?	K/ CO1
3.	Explain UV-Visible absorption.	U / CO2
4.	What is the information derived from IR spectrum.	U / CO2
5.	Discuss the origin of Raman spectra.	U / CO2
6.	What are the different interactions of electron beam with a sample.	K / CO3
7.	Describe Transmission Electron Microscopy.	U / CO3
8.	Mention the key inferences from an energy dispersive spectrum.	An / CO3
9.	Describe tunneling current in Scanning Tunneling Microscopy.	K / CO3
10.	Discuss the different near-field forces in Atomic Force Microscopy.	K / CO3
11.	What do you mean by thermal events and thermal analysis?	K / CO4
12.	What can you describe about the sample holder and reference material used in the instrumentation of thermal analysis?	U, A, An / CO4

13	What do you mean by modulated DSC?	U, A, An / CO4
14.	What are the main objectives of thermogravimetry study?	An / CO4

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

15.	How to determine the crystallite size and lattice parameters of a crystal.	U, A / CO1
16.	Describe the experimental setup for Raman scattering.	U / CO2
17.	Discuss the basic principles of Photoluminescence.	U / CO2
18.	Explain the principle and instrumentation of Scanning Electron Microscopy.	U / CO3
19.	Discuss the instrumentation of Scanning Probe Microscopy.	K / CO3
20.	What are the four operational modes in Scanning Tunnelling Microscopy.	U / CO3
21.	Discuss the working principle if Differential Thermal Analysis (DTA).	U / CO4
22.	Describe the instrumentation thermogravimetry.	K / CO4

[5x6 = 30]